

JOSHUA BINU KOCHIYIL

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CAREER OBJECTIVE

Advanced Computing (Honours) graduate with hands-on experience developing full-stack applications and applied AI and LLM-based systems across production environments. Experienced in developing React-based interfaces, backend services, and operational ML pipelines with a focus on reliability, scalability, and real-world impact. Seeking to contribute strong technical execution and ownership while continuing to grow within a collaborative team building impactful software systems.

EDUCATION

Bachelor of Advanced Computing (Honours)
The Australian National University

Canberra, Australia
2021 - 2025

- **Specialisation: Machine Learning**
- **GPA: 6.31 / 7.00**
- Recipient of **ANU Chancellor's International Scholarship**
- **Relevant Coursework:** Software Engineering, Natural Language Processing, Computer Vision, Statistical Machine Learning, Data Structures and Algorithms, Database Management, Systems & Networks Programming

TECHNICAL SKILLS

Programming & Automation: Python, Java, C/C++, JavaScript/TypeScript, SQL, Selenium

ML & AI: PyTorch, TensorFlow, Scikit-Learn, SHAP, NLP, Explainable AI (XAI), OCR, Gen AI

Databases: MySQL, PostgreSQL, Firebase, Supabase

Web & Frontend: React, Remix, HTML/CSS, Tailwind CSS, Flutter, Figma

Backend & APIs: Flask, FastAPI, Django, API Development

Cloud & DevOps: Microsoft Azure, Docker, CI/CD (GitHub Actions), Git, Unix/Linux

Engineering Practices: Agile Scrum, User Story Mapping, Test-Driven Development, Technical Documentation

WORK EXPERIENCE

Machine Learning Research Assistant Intern
ANU Research School of Biology

Canberra, Australia
Nov 2024 - Present

- Designed and deployed an **LLM-enhanced SHAP optimisation pipeline** in Python, reducing computation cost by **20%** while maintaining reliable outputs.
- Developed a **web-based dashboard** that integrated a **conversational AI interface**, enabling non-technical users to explore machine learning model outputs through natural language queries and interactive visualisations.
- Implemented a **Retrieval-Augmented Generation (RAG)** pipeline with optimised **prompt engineering** to ground LLM responses, reducing hallucinations and improving response consistency.
- Built **containerised microservices using Docker** to support reproducible builds and seamless cloud integration.
- Collaborated with stakeholders to define **success metrics and usability criteria**, ensuring outputs aligned with real-world decision-making needs.

Full-Stack Software Engineer
Ohna

Canberra, Australia
Feb 2024 - Mar 2025

- Led the design and development of a **real-time logistics automation system** (Remix + Supabase), improving packing throughput by **15%** across **500+ weekly orders**.
- Built and deployed **Python & PostgreSQL automation pipelines** for a high-volume operational database, maintaining data quality and purging stale records to reduce manual overhead and improve query performance.
- Designed **secure relational schemas** in Supabase, implementing **row-level security (RLS)** and role-based access control aligned with cloud security best practices.

- Maintained and optimised a production **Flutter-Firebase e-commerce mobile app** by identifying and resolving critical bugs, improving overall app stability and user experience.
- Secured a **\$30,000 Innovation Connect Grant** from the Canberra Innovation Network by delivering **live technical demonstrations** and presenting the platform's **community and operational value**.
- Led end-to-end delivery within an **Agile Scrum** environment, contributing to **sprint planning** and feature development through structured Git workflows.

OTHER EXPERIENCE

Casual Sales Assistant

Shaver Shop

Canberra, Australia

Jul 2025 - Present

- **Customer Engagement & Needs Analysis:** Consistently exceeded sales KPIs through effective customer needs analysis and tailored product recommendations, demonstrating strong communication and active listening.

Casual Retail Team Member

JB Hi-Fi

Canberra, Australia

Jan 2024 - Dec 2024

- **Problem Solving:** Resolved customer complaints and defective product cases through calm, solution-oriented communication.

SOFT SKILLS

Stakeholder Communication: Collaborated with cross-functional teams and non-technical stakeholders to understand requirements, translate them into practical system solutions, and provide ongoing user support.

Problem Solving & Continuous Improvement: Analysed operational challenges, identified inefficiencies, and implemented practical solutions that improved workflows, reduced manual effort, and enhanced system performance.

Execution & Time Management: Delivered multiple core system components across parallel sprints in a fast-paced startup environment while consistently meeting project deadlines.

PERSONAL PROJECTS

AI Expense Tracker

Tech Stack: Remix, FastAPI, PostgreSQL, OCR, LLM Integration, Docker, GitHub Actions, Selenium

- Built a full-stack **AI-powered expense analytics platform** with a multi-stage pipeline to extract and structure data from receipt images using **OCR** and **LLMs**.
- Developed a **natural language-to-SQL query engine** to enable intuitive querying of financial data.
- Designed interactive dashboards to **visualise spending patterns, trends, and anomalies** for decision-making.
- Implemented **end-to-end automated testing using Selenium** to validate the full user workflow, ensuring reliable performance across receipt upload, AI data extraction, query processing, and dashboard functionality.

MADI (Medical Analysis & Diagnostics Intelligence)

Tech Stack: Python, Flask, Scikit-Learn, REST APIs, Pytest

- Built and deployed a **Flask backend** that unified multiple disease prediction models into a single platform, aiming to support medical professionals in reaching **faster and more accurate diagnoses**.
- Implemented **unit and integration testing using Pytest** for ML inference endpoints to ensure dependable data flows and prediction accuracy.
- Built a **custom AI chatbot** that interprets model outputs and generates patient-friendly explanations of diagnostic predictions, improving accessibility of complex medical insights.

ACHIEVEMENTS

- **McCusker Prize in Humanitarian Engineering (2024):** Awarded for outstanding real-world impact of project *Ohna* in tackling food insecurity through technology, demonstrating engineering excellence with humanitarian value.
- **ANUEC Hackathon Finalist (2024):** Advanced to the final round in the ANU Entrepreneurship Club (ANUEC) 24-hour hackathon with project *MADI*. Demonstrated the project's technical depth and social impact to a panel of industry experts, highlighting innovation in healthcare accessibility and technical execution.